



# Pairwise Intent Graph Embedding Learning for Context-Aware Recommendation

RecSys 23

# Contents

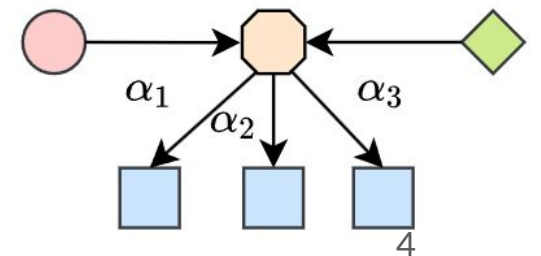
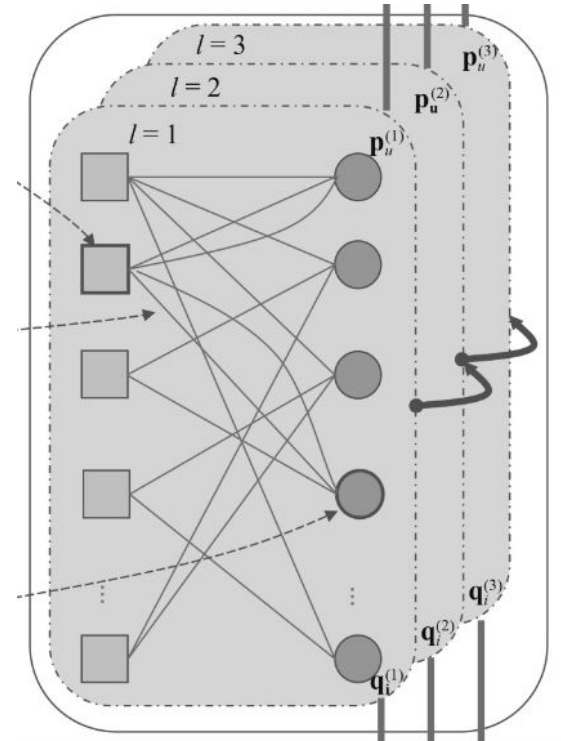
- Background
- Preliminaries
- Methods
- Experiments
- Conclusion & Research Idea

---

# Background

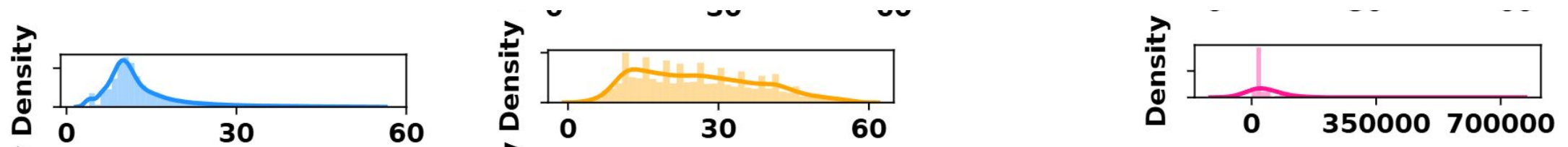
# Context-Aware Recommender Systems (CARS)

- Recommender system that consider contextual information
  - e.g. city, year, month and day of the week
- Neural network-based methods
- GCM: user-item bipartite graph
  - Contextual features are used as edges between users and items
  - Contextual features do not benefit from graph convolution
- UEG: user-event graph structure
  - Intents driven by contextual information act as central nodes



# KG-Based Recommender Systems

- Knowledge Graph (KG) has been widely used in recommendation tasks
  - Propagation-based methods: iterative aggregation of information from neighbors
  - learning of high-order relationships in end-to-end manner
- KGs are effective in data sparsity
  - It might be beneficial for two data sparsity issues in CARS
  - **feature sparsity**: long-tailed distribution of contextual features
  - **interaction sparsity**: most contextual features occur infrequently



feature sparsity (left: users, right: items)

interaction sparsity



# Preliminaries

# Problem Definition

- Context-Aware Recommender Systems (CARS)

$$U = \{u_1, u_2, \dots, u_M\}, I = \{i_1, i_2, \dots, i_N\}, C = \{C_1, C_2, \dots, C_R\}$$

$$s_k = [u^k, i^k, C^k] \text{ where } u^k \in U, i^k \in I, C^k \subset C$$

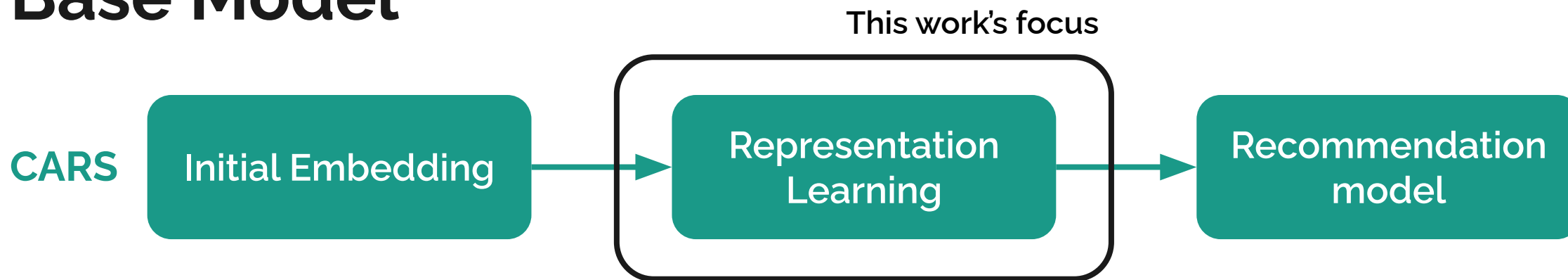
- Knowledge Graph (KG)

- $G_{kg} = \{(h, r, t) \mid h, t \in V, r \in R\}$
- We assume KG associated with the items is available

- Goal

- Leverage KG as auxiliary data source
- Accurately predict item that is most likely to be interacted by a user under a context

# Base Model



- Initial Embedding
  - Sparse, high-dimensional vector  $\rightarrow$  Dense, low-dimensional real-valued vector
- Recommendation model
  - Predicted score  $\hat{y}(s_k) = \sigma(b_g + \sum b_{\star} + \frac{1}{2}[(\sum \mathbf{e}_{\star})^2 - \sum \mathbf{e}_{\star}^{\top} \mathbf{e}_{\star}])$   $\star \in \{u^k, i^k, C^k\}$
  - Objective function (BCE loss)

$$\mathcal{L} = -\frac{1}{|S'|} \sum_{(s_i, y) \in S'} y_i \log \hat{y}(s_i) + (1 - y_i) \log(1 - \hat{y}(s_i))$$

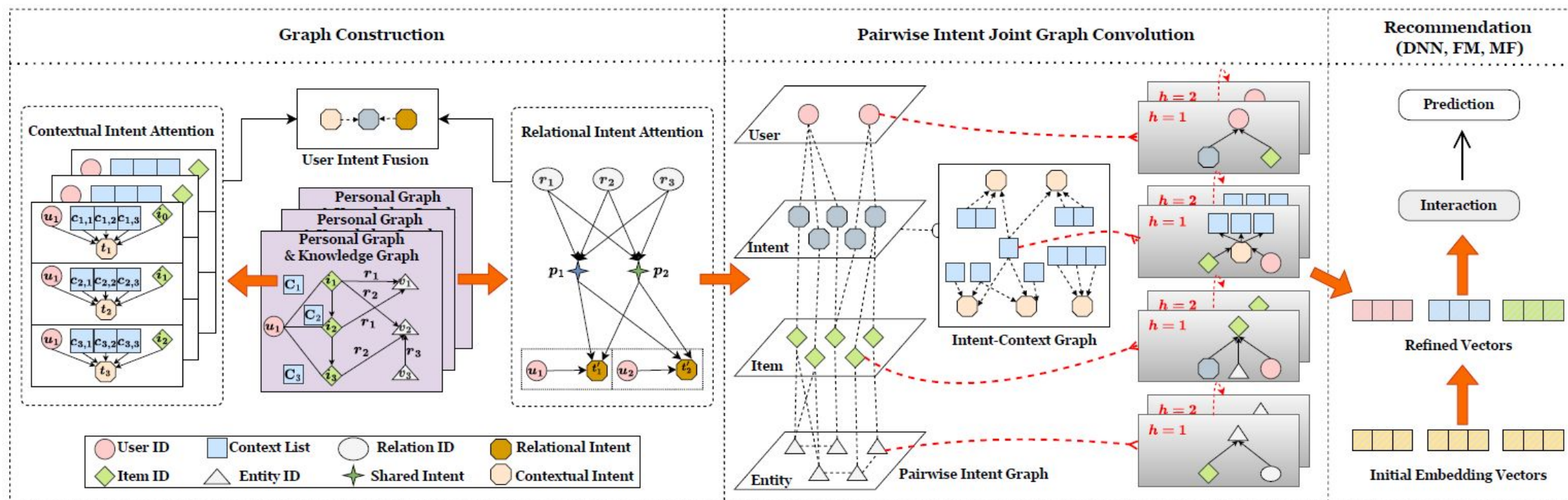


---

# Methods

# Overview

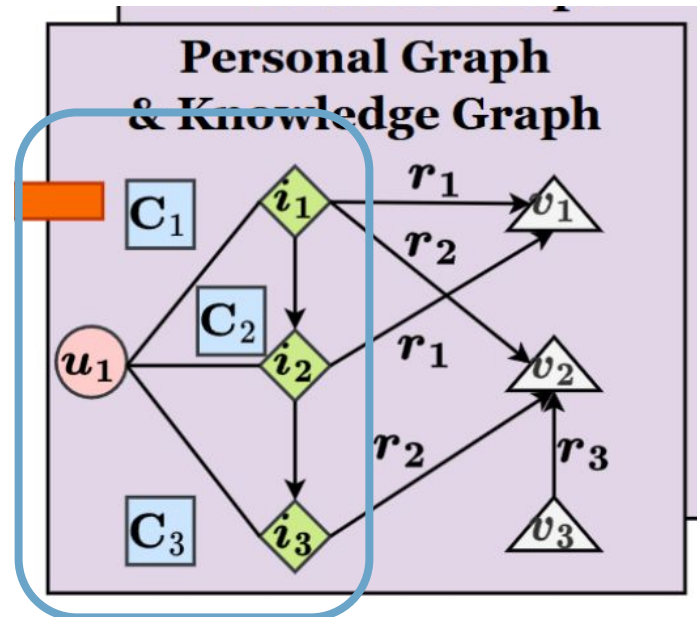
## ● PING (Pairwise Intent Graph Embedding Learning) framework



# Graph Construction

- $G_{pg} = \langle V_{pg}, E_{pg} \rangle$  where  $V_{pg} = \{u\} \cup I$  and  $E_{pg} = E_{ui} \cup E_{ii}$
- $G_{pg\&kg} = \langle V_{pg\&kg}, E_{pg\&kg} \rangle$

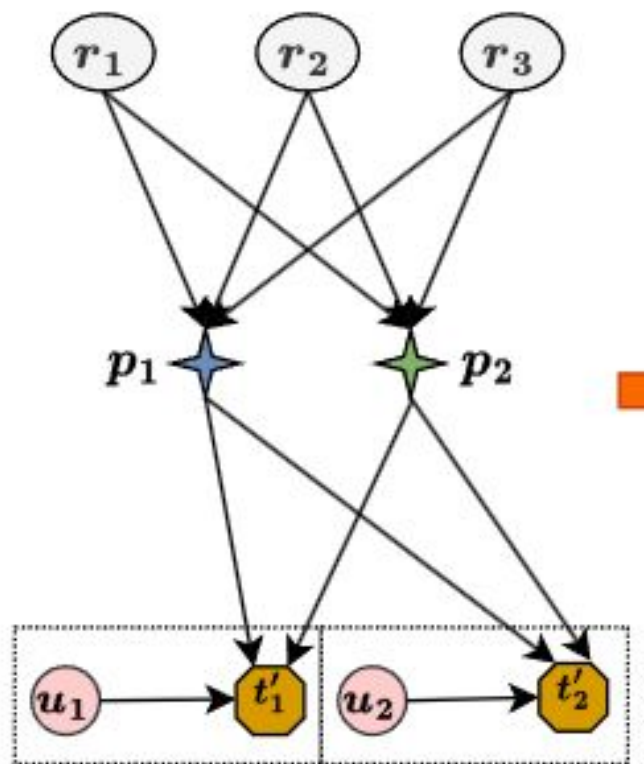
where  $V_{pg\&kg} = \{u\} \cup I \cup V$  and  $E_{pg\&kg} = E_{ui} \cup E_{ii} \cup E_{iv} \cup E_{vv}$



Personal Graph

# Graph Construction

## Relational Intent Attention



Calculation of the shared intent

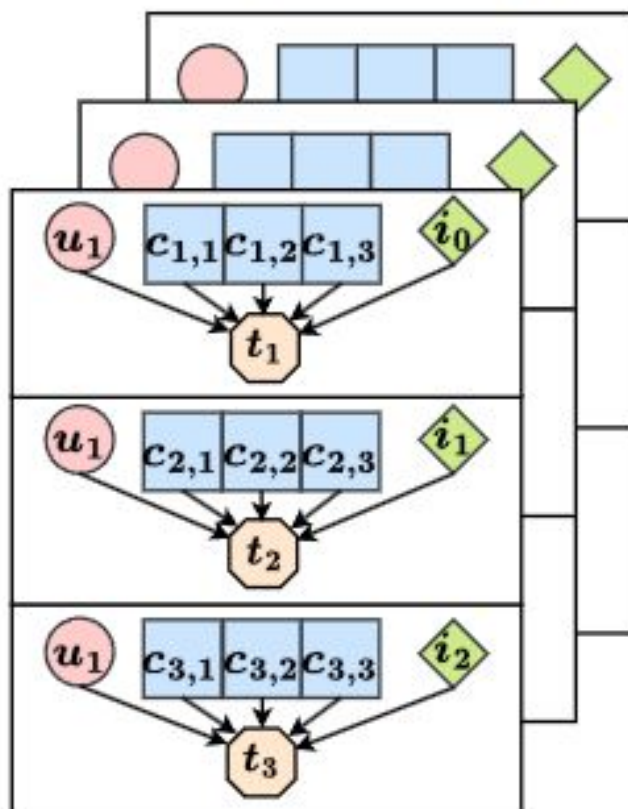
$$\mathbf{e}_p = \sum_{r \in \mathcal{R}} \alpha_{rp} \mathbf{e}_r,$$
$$\alpha_{rp} = \frac{\exp(w_{rp})}{\sum_{r' \in \mathcal{R}} \exp(w_{r'p})}$$

Calculation of the relational intent

$$\mathbf{e}_{t'_u} = \sum_{p \in \mathcal{P}} \beta_{up} \mathbf{e}_p,$$
$$\beta_{up} = \frac{\exp(\mathbf{e}_p^\top \mathbf{e}_u)}{\sum_{p' \in \mathcal{P}} \exp(\mathbf{e}_{p'}^\top \mathbf{e}_u)}$$

# Graph Construction

## Contextual Intent Attention



Calculation of the contextual intent

$$\gamma_z^k = \text{Softmax}(\mathbf{W}_0^\top \text{Relu}(\mathbf{W}_1 \mathbf{e}_{u^k} + \mathbf{W}_2 \mathbf{e}_{c_z^k} + \mathbf{b}_1)),$$

$$\mathbf{e}_{t^k} = \sum_{z=1}^{Z+1} \gamma_z^k \mathbf{e}_{c_z^k},$$

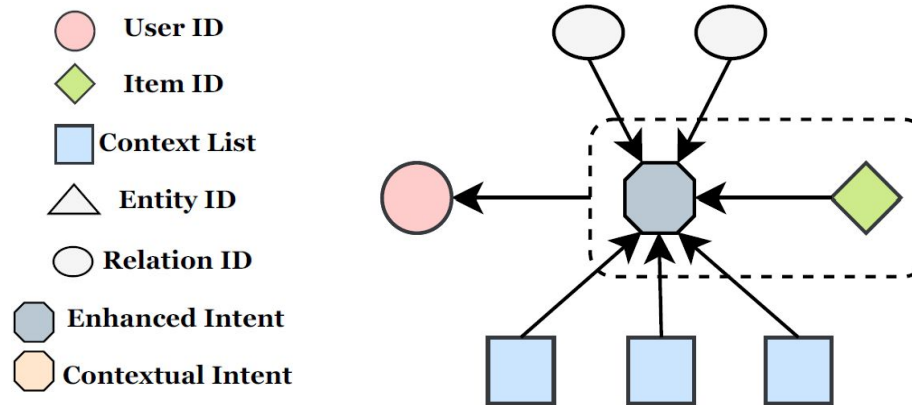
Final fusion (enhanced intent node)

$$\mathbf{e}_{\bar{t}^k} = \frac{(\mathbf{e}_{t'_u} + \mathbf{e}_{t^k})}{2}$$

# Pairwise Intent Joint Graph Convolution

- Information Propagation for the Users

- Capture the **item and contextual** information **for a user**.
- Utilize **an enhanced intent node** as the hub to complete the desired information propagation.



# Pairwise Intent Joint Graph Convolution

## ● Information Propagation for the Users

- Information representation **passed to the user**

$$\mathbf{p}_{u^k, k}^{(h)} = \mathbf{p}_{i^k}^{(h-1)} + \mathbf{p}_{\bar{i}^k}^{(h-1)}$$

- **Aggregation** to obtain each user's embedding at layer  $h$

$$\mathbf{p}_u^{(h)} = \frac{1}{\sqrt{|\{k | u^k = u\}|}} \sum_{k, u^k = u} \mathbf{p}_{u, k}^{(h)}$$

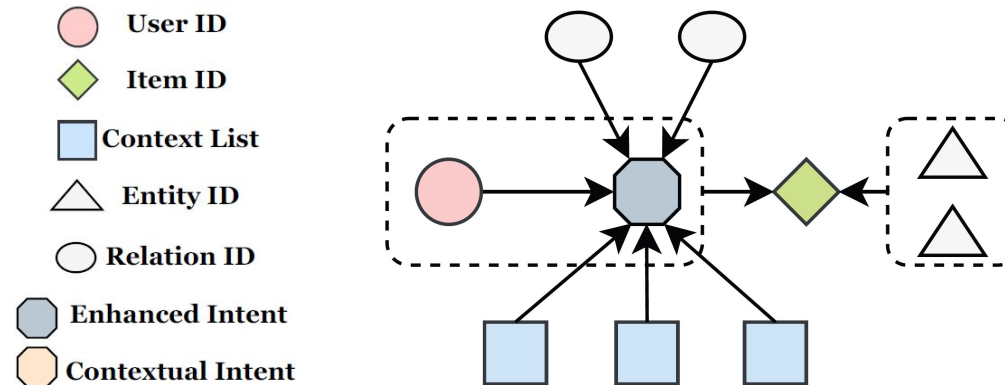
- **Refined user embedding**

$$\hat{\mathbf{p}}_u = \frac{1}{H+1} \sum_{h=0}^H \mathbf{p}_u^{(h)}$$

# Pairwise Intent Joint Graph Convolution

## ● Information Propagation for the Items

- Capture the **user and contextual** information **for a item**.
- Utilize **an enhanced intent node** as the hub to complete the desired information propagation.





# Pairwise Intent Joint Graph Convolution

## ● Information Propagation for the Items

- Information representation **passed to the item**

$$\mathbf{p}_{i^k,k}^{(h)} = \mathbf{p}_{u^k}^{(h-1)} + \mathbf{p}_{\bar{t}^k}^{(h-1)}$$

- **Aggregation** to obtain each **item's embedding** at layer  $h$

$$\mathbf{p}_i^{(h)} = \frac{1}{\sqrt{|\{k|i^k = i\}|}} \sum_{k,i^k=i} \mathbf{p}_{i,k}^{(h)} + \frac{1}{|N_i|} \sum_{(r,v) \in N_i} \mathbf{e}_r \odot \mathbf{p}_v^{(h-1)}$$

where  $N_i$  denotes the set of relations and entities associated with item  $i$

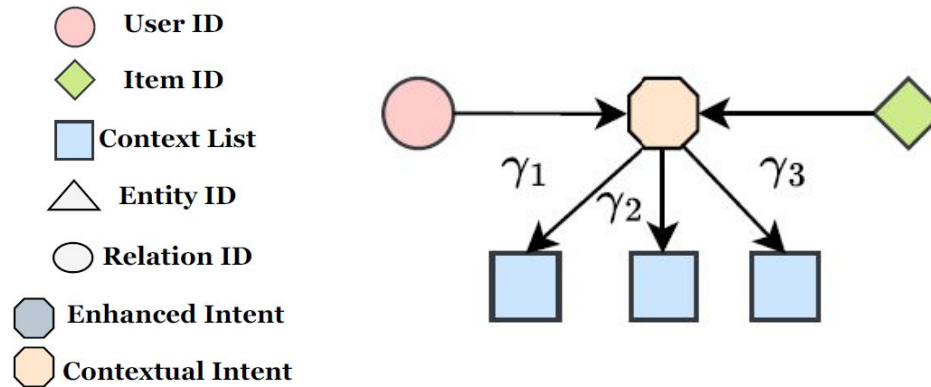
- **Refined item embedding**

$$\hat{\mathbf{p}}_i = \frac{1}{H+1} \sum_{h=0}^H \mathbf{p}_i^{(h)}$$

# Pairwise Intent Joint Graph Convolution

## ● Information Propagation for the Context

- First, propagate the **user and item** information **to intent nodes**.
- Then, **distribute** the information of contextual features according to the **contextual attention**.



# Pairwise Intent Joint Graph Convolution

## ● Information Propagation for the Context

- Propagation of user and item information

$$\mathbf{p}_{t^k,k}^{(h)} = \mathbf{p}_{u^k}^{(h-1)} + \mathbf{p}_{i^k}^{(h-1)} \qquad \mathbf{p}_{c_z^k,k}^{(h)} = \gamma_z^k \mathbf{p}_{t^k,k}^{(h)}$$

- **Refined embedding** of the context

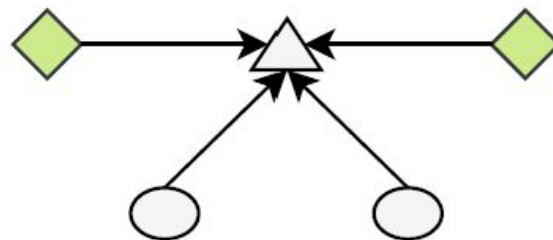
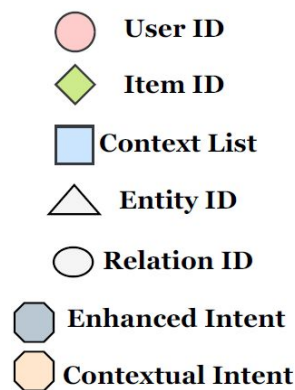
$$\mathbf{p}_{c_z}^{(h)} = \frac{1}{\sqrt{|\{k | c_z^k = c_z\}|}} \sum_{k, c_z^k = c_z} \mathbf{p}_{c_z^k,k}^{(h)} \qquad \hat{\mathbf{p}}_{c_z} = \frac{1}{H+1} \sum_{h=0}^H \mathbf{p}_{c_z}^{(h)}$$

- Embedding of contextual intent

$$\mathbf{p}_{t^k}^{(h)} = \sum \gamma_z^k \mathbf{p}_{c_z^k}^{(h)}$$

# Pairwise Intent Joint Graph Convolution

## ● Information Propagation for the Entity



$$\hat{\mathbf{p}}_v^{(h)} = \frac{1}{|\mathcal{N}_v|} \sum_{(r,i) \in \mathcal{N}_v} \mathbf{e}_r \odot \mathbf{p}_i^{(h-1)}$$

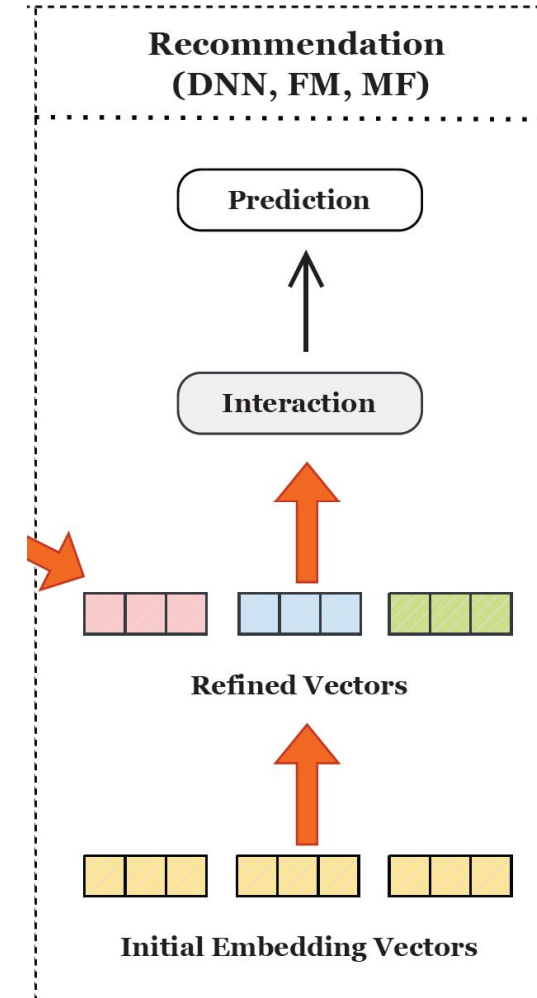
where  $\mathcal{N}_v$  denotes the set of relations and items associated with entity  $v$

# Recommendation Module

## ● Refined Embeddings

- Use refined embeddings instead of initial embeddings
- **Compatible** with various downstream recommendation models

$$\mathbf{P}_{S_k} = [\hat{\mathbf{p}}_u^k \quad \hat{\mathbf{p}}_i^k \quad \hat{\mathbf{p}}_C^k]$$



---

# Experiments

# Experimental Setup

## ● Datasets

- **Amazon-book**: a subset corresponding to the book product category in Amazon dataset.
- **Yelp2018**: extensive records of user interactions with local businesses
- **Last-FM**: a music listening dataset collected from the online music platform.

**Table 1: Statistics of the processed datasets.**

|                          |                     | Amazon-book | Yelp2018  | Last-FM   |
|--------------------------|---------------------|-------------|-----------|-----------|
| User-Item<br>Interaction | #Users              | 70,679      | 23,566    | 45,919    |
|                          | #Items              | 24,915      | 48,123    | 45,538    |
|                          | #Interactions       | 847,733     | 3,034,796 | 1,185,068 |
|                          | #Contextual Feature | 59          | 535       | 133       |
| Knowledge<br>Graph       | #Entities           | 88,572      | 58,266    | 90,961    |
|                          | #Relations          | 39          | 9         | 42        |
|                          | #Triplets           | 2,557,746   | 464,567   | 1,853,704 |

# Experimental Setup

## ● Datasets Preprocessing

### ○ Amazon-book & Last-FM

- Map items to entities into Freebase knowledge graph via title mapping, if available.
- Contextual features: year, month, day, and day\_of\_week

### ○ Yelp2018

- Extract item knowledge from the local business information network
- Contextual features: city, year, month, day\_of\_week

### ○ Training: 70%, Validation: 10%, Test: 20%



# Experimental Setup

- **Baselines**

- Context-aware methods: FM, GCM, and UEG
- Knowledge graph-based methods: CKE, KGNN-LS, CKAN, KGAT, KGIN, and HAKG
- Classical Recommendation method: MF

- **Evaluation Metrics**

- Recall @ 20, NDCG @ 20
- Average for all users in the testing set

# Performance Comparison

|               | Dataset     | Amazon-book    |                | Yelp2018       |                | Last-FM        |                |
|---------------|-------------|----------------|----------------|----------------|----------------|----------------|----------------|
|               | Metrics     | R@20           | N@20           | R@20           | N@20           | R@20           | N@20           |
| KG-based      | MF          | 0.1300         | 0.0678         | 0.0627         | 0.0413         | 0.0724         | 0.0617         |
|               | CKE         | 0.1342         | 0.0698         | 0.0653         | 0.0423         | 0.0732         | 0.0630         |
|               | KGNN-LS     | 0.1362         | 0.0560         | 0.0671         | 0.0422         | 0.0880         | 0.0642         |
|               | CKAN        | 0.1442         | 0.0698         | 0.0646         | 0.0441         | 0.0812         | 0.0660         |
|               | KGAT        | 0.1487         | 0.0799         | 0.0705         | 0.0463         | 0.0873         | 0.0744         |
|               | KGIN        | 0.1687         | 0.0915         | 0.0698         | 0.0451         | 0.0978         | 0.0848         |
|               | HAKG        | 0.1421         | 0.0863         | 0.0778         | 0.0501         | 0.1008         | <u>0.0931</u>  |
| Context-aware | FM          | 0.1565         | 0.0793         | 0.1186         | 0.0550         | 0.1084         | 0.0433         |
|               | GCM (w/o c) | 0.1371         | 0.0618         | 0.0761         | 0.0309         | 0.0839         | 0.0387         |
|               | GCM         | 0.1863         | 0.0919         | 0.1285         | 0.0556         | <u>0.1088</u>  | 0.0496         |
|               | UEG (w/o c) | 0.1399         | 0.0623         | 0.0765         | 0.0317         | 0.0747         | 0.0346         |
|               | UEG         | <u>0.2872</u>  | <u>0.1583</u>  | <u>0.1352</u>  | <u>0.0613</u>  | 0.0940         | 0.0441         |
|               | PING        | <b>0.3043*</b> | <b>0.1759*</b> | <b>0.2132*</b> | <b>0.1336*</b> | <b>0.1752*</b> | <b>0.1014*</b> |

# Ablation Study

## ● Ablation Study

- RI: Relational Intent
- CI: Contextual Intent

| Dataset   | Amazon-book   |               | Yelp2018      |               | Last-FM       |               |
|-----------|---------------|---------------|---------------|---------------|---------------|---------------|
| Metrics   | R@20          | N@20          | R@20          | N@20          | R@20          | N@20          |
| PING      | <b>0.3043</b> | <b>0.1759</b> | <b>0.2132</b> | <b>0.1336</b> | <b>0.1752</b> | <b>0.1014</b> |
| w/o RI    | 0.2714        | 0.1546        | 0.1677        | 0.0886        | 0.1459        | 0.0731        |
| w/o CI    | 0.2537        | 0.1254        | 0.1521        | 0.0715        | 0.1439        | 0.0673        |
| w/o RI&CI | 0.1863        | 0.0919        | 0.1285        | 0.0556        | 0.1088        | 0.0496        |

# Interpretability of the framework

Context in Each Intent

| Context       | Score |
|---------------|-------|
| $c_1$ (year)  | 0.105 |
| $c_2$ (month) | 0.078 |
| $c_3$ (day)   | 0.092 |
| $c_4$ (DoW)   | 0.024 |
| $c_5$ (last)  | 0.701 |

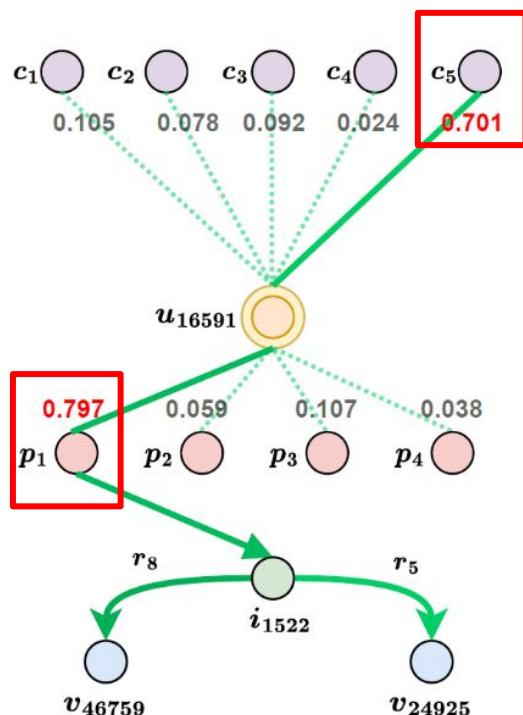
Top 2 KG Relations in Each Intent

| Intent | Relation                           | Score |
|--------|------------------------------------|-------|
| $p_1$  | $r_8$ (book.book_subject.works)    | 0.612 |
|        | $r_5$ (book.written_work.subjects) | 0.137 |

|       |                                      |       |
|-------|--------------------------------------|-------|
| $p_2$ | $r_{14}$ (book.author.works_written) | 0.386 |
|       | $r_{19}$ (book.book.characters)      | 0.168 |

|       |                                   |       |
|-------|-----------------------------------|-------|
| $p_3$ | $r_{24}$ (book.short_story.genre) | 0.144 |
|       | $r_8$ (book.book_subject.works)   | 0.132 |

|       |                                           |       |
|-------|-------------------------------------------|-------|
| $p_4$ | $r_{36}$ (book.interior_illustrations_by) | 0.327 |
|       | $r_{10}$ (book.written_work.author)       | 0.262 |



Context in Each Intent

| Context       | Score |
|---------------|-------|
| $c_1$ (city)  | 0.425 |
| $c_2$ (year)  | 0.108 |
| $c_3$ (month) | 0.063 |
| $c_4$ (DoW)   | 0.215 |
| $c_5$ (last)  | 0.189 |

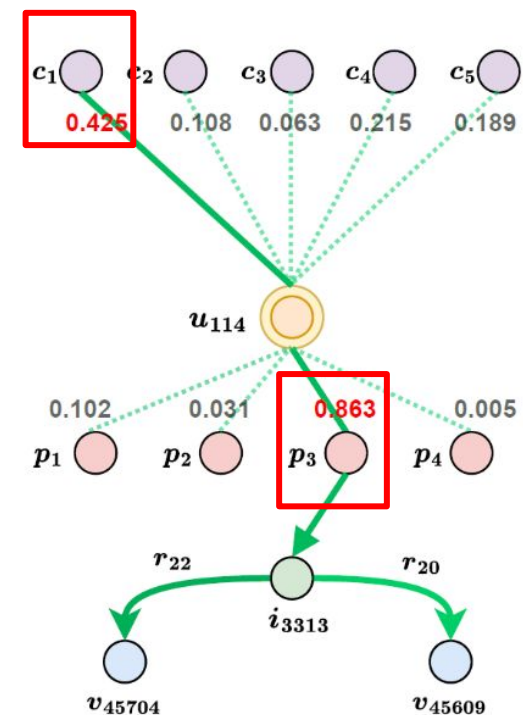
Top 2 KG Relations in Each Intent

|       |                                    |       |
|-------|------------------------------------|-------|
| $p_1$ | $r_{27}$ (Music)                   | 0.515 |
|       | $r_{22}$ (RestaurantsTableService) | 0.163 |

|       |                        |       |
|-------|------------------------|-------|
| $p_2$ | $r_{35}$ (BYOBCorkage) | 0.303 |
|       | $r_5$ (OutdoorSeating) | 0.129 |

|       |                                    |       |
|-------|------------------------------------|-------|
| $p_3$ | $r_{22}$ (RestaurantsTableService) | 0.316 |
|       | $r_{20}$ (star)                    | 0.272 |

|       |                        |       |
|-------|------------------------|-------|
| $p_4$ | $r_5$ (OutdoorSeating) | 0.254 |
|       | $r_{28}$ (Smoking)     | 0.166 |

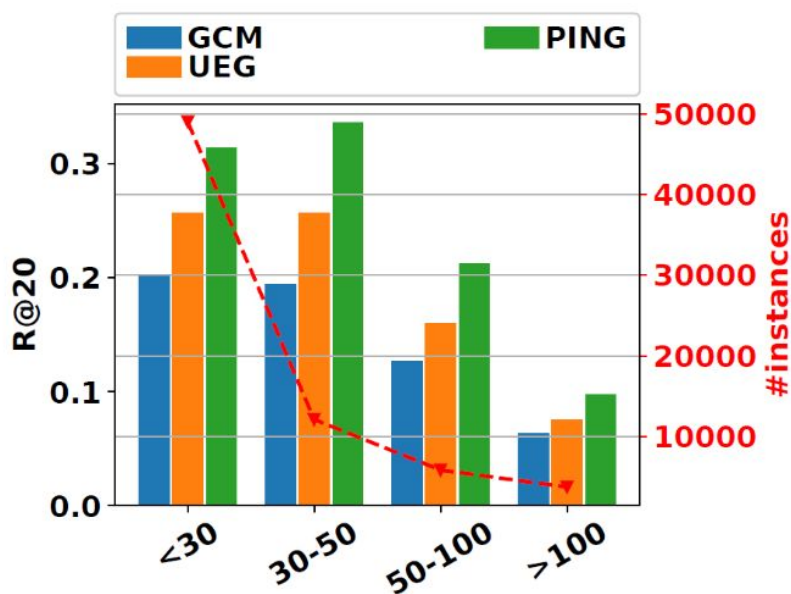




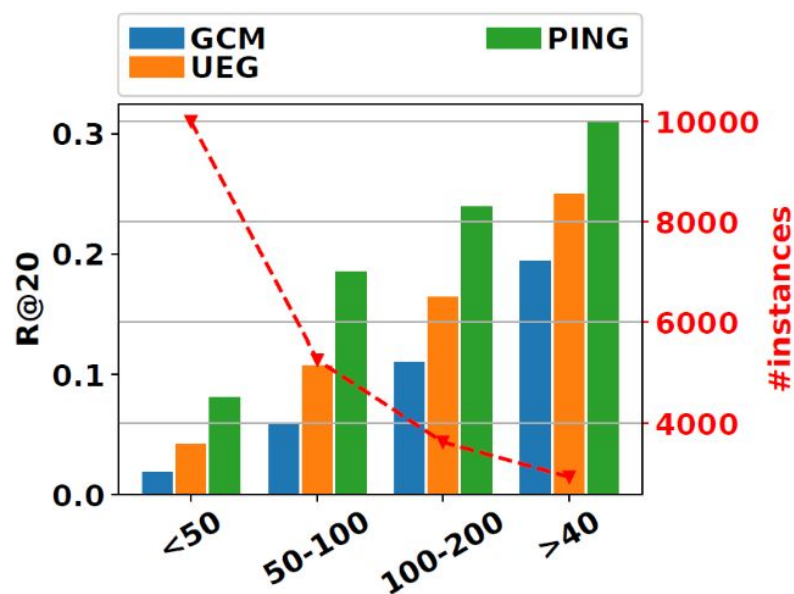
# Feature & Interaction Sparsity

- PING improves feature sparsity and interaction sparsity problems.

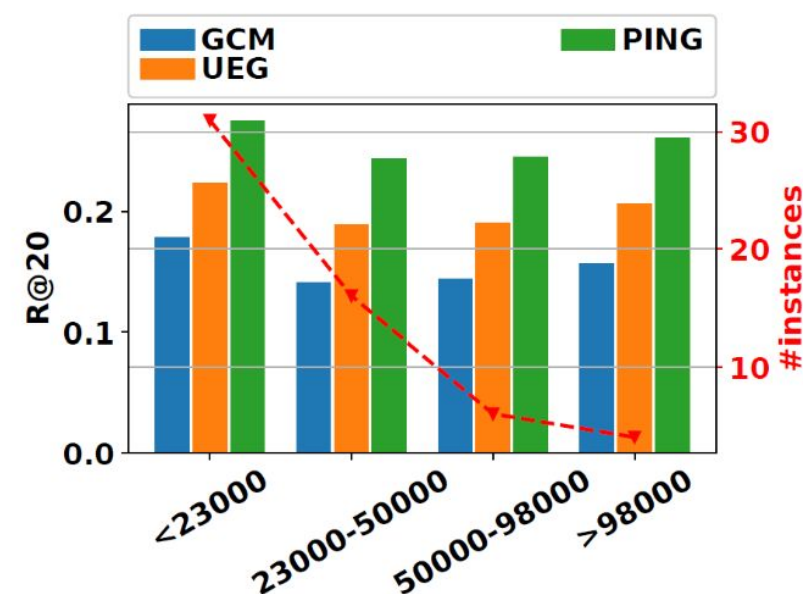
- with Amazon-book dataset



(a) u-c interaction



(b) i-c interaction



(c) context frequency

---

# Conclusion & Research Idea

# Conclusion

- Pairwise intent graph embedding learning framework
  - Utilize knowledge graphs to combine high-order information of users, items, and contexts
  - Mitigate two data sparsity problems: **feature sparsity and interaction sparsity**
  - Three modules:
    - **Graph Construction Module**
    - **Pairwise Intent Joint Graph Convolution Module**
    - **Recommendation Module**

# Research Idea

- Utilize KG to extract semantic information of users and contexts.
  - PING combine entities of a KG with items via title mapping.
    - **Not always guarantee** that each item is involved in the KGs.
  - Context information can be augmented with KGs.
    - **Location**, Time (**Month**, **Day of Week**)
  - New datasets, graph construction, and convolution mechanism are required.
    - Time: **numeric values**



# Research Idea

- Considering inductive bias of dependency between interaction records from same user.
  - PING **only aggregates associated instances** to compute embeddings of users, items, and contextual features.
    - It failed to utilize dependencies between instances associated with a same user
  - Introducing inductive bias to **learn dependency** might be helpful for **sequential recommendation**.
    - Capture **dependencies** between interactions via **enhanced intent**.
    - Also, it can improve **interpretability** of the framework.

# Q&A

---